Nonnegative and positive factorizations

Exact positive semidefinite rank of subset-intersection matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: Open

Rating rationale: Challenging because the symmetric subset structure has not yielded matching PSD-rank bounds; specialist importance reflects this explicit family's role as a benchmark for semidefinite factorization.

Area: structured positive semidefinite factorization

Context and notation

Write $\mathbb{S}_+^k$ for the cone of real symmetric positive semidefinite $k \times k$ matrices. For an entrywise nonnegative matrix $M \in \mathbb{R}_+^{p\times q}$, its **real positive semidefinite rank** is

$$\operatorname{rank}_{\mathrm{psd}}(M) = \min\{k \geq 1 : \exists A_1, \dots, A_p, B_1, \dots, B_q \in \mathbb{S}_+^k, \quad M_{ij} = \operatorname{tr}(A_i B_j) \text{ for all } i, j\}.$$

This definition concerns a family of matrix factors indexed by the rows and columns of M.

Problem statement

For each integer $n \geq 5$, let $\mathcal{I}_n$ and $\mathcal{J}_n$ consist of the subsets of $\{1, \dots, n\}$ of cardinalities $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$, respectively. Define the nonnegative matrix

$$M^{(n)} \in \mathbb{R}_+^{\mathcal{I}_n \times \mathcal{J}_n}, \qquad M_{I,J}^{(n)} = |I \cap J|.$$

Question

Determine $\operatorname{rank}_{\mathrm{psd}}(M^{(n)})$ exactly as a function of n, with real symmetric factors as defined above. The family is one problem; its individual orders are not separate catalog entries.

The source supplies the bounds

$$\left\lceil \frac{\sqrt{1+8n}-1}{2} \right\rceil \leq \operatorname{rank}_{\mathrm{psd}}(M^{(n)}) \leq 2\lceil \sqrt{n} \rceil.$$

These matrices provide structured benchmarks for algorithms seeking small positive semidefinite factorizations.

References

Hamza Fawzi, João Gouveia, Pablo A. Parrilo, Richard Z. Robinson, and Rekha R. Thomas, *Positive semidefinite rank*, Mathematical Programming **153** (2015), 133–177, §9.1, Problem 9.2. Arnaud Vandaele, François Glineur, and Nicolas Gillis, *Algorithms for Positive Semidefinite Factorization*, Computational Optimization and Applications **71** (2018), 193–219, §4.2, preprint pp. 10–11.

Status check — 2026-09-10

Rechecked Fawzi et al., Problem 9.2 and Vandaele–Glineur–Gillis, §4.2, and searched by subset intersections, Johnson schemes and the problem number. The sources support the displayed real-factor bounds but not an exact formula. No later determination was located; the 2018 algorithm paper remains the latest explicit status source checked, so the later evidence is limited to this search.